PROJECT DOCUMENTATION

EV MARKET ANALYSIS DASHBOARD

Using Tableau Public

Dataset: ev_market_train.csv

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1. INTRODUCTION

1.1 Project Overview

Electric vehicles (EVs) are an important part of the transition towards sustainable transportation. EVs differ in battery technology, driving range, charging performance, price, manufacturer, safety ratings, and other technical specifications. Understanding these characteristics can help users explore the electric vehicle market and compare vehicles according to their requirements.

The EV Market Analysis Dashboard is a data visualisation project developed around the ev_market_train.csv dataset. The previously inspected version contains 200,000 records and 37 columns. The dataset includes information about vehicle manufacturers, model years, manufacturing countries, market regions, vehicle categories, battery specifications, driving range, energy efficiency, motor performance, charging characteristics, pricing, insurance costs, warranties, safety ratings, customer ratings, annual sales, and estimated CO_2 savings.

The primary objective is to convert a large and complex dataset into understandable visual information using Tableau Public. Instead of manually examining thousands of vehicle records, users can explore summaries, compare categories, apply filters, and investigate differences among manufacturers and vehicle specifications.

The project focuses on descriptive data analysis and visualisation. It is not intended to predict future EV sales or establish causal relationships between vehicle characteristics.

1.2 Purpose

The main purpose of the project is to make EV market data easier to understand through interactive visualisations.

1. Organise and present EV market information in a clear dashboard.
2. Compare recorded annual sales across manufacturers and vehicle categories.
3. Explore battery technologies, battery capacity, driving range, and energy efficiency.
4. Investigate vehicle prices in relation to driving range.
5. Compare customer ratings and safety ratings across vehicles or manufacturers.
6. Examine manufacturing countries, market regions, and model years.
7. Help users identify patterns and differences in the supplied dataset.
8. Support evidence-based observations through charts, filters, and summary indicators.

The project is intended for students, researchers, EV enthusiasts, and analysts who need an accessible way to explore the supplied EV dataset.

2. IDEATION PHASE

2.1 Problem Statement

The EV dataset contains numerous records and many different attributes. Although these fields provide useful information, interpreting them through a raw spreadsheet can be time-consuming. Users may need to switch between columns, apply manual filters, calculate summaries, and compare multiple groups before understanding the patterns.

The central problem is that large amounts of EV market data are difficult to compare efficiently when presented only in tabular form.

Customer Problem Statement: I am an EV market analyst or student trying to compare manufacturers, vehicle categories, battery technologies, driving ranges, prices, and customer ratings. However, the information is spread across a large number of records and attributes. Manually comparing these values makes analysis time-consuming and makes important patterns harder to identify. Therefore, I need an interactive dashboard that presents the major comparisons through clear visualisations and useful filters.

Project Problem Statement: To design an interactive EV Market Analysis Dashboard using Tableau Public that transforms the supplied EV dataset into clear, filterable visual summaries of sales, technical specifications, pricing, ratings, geography, and sustainability indicators.

2.2 Empathy Map Canvas

The empathy map identifies the potential needs and difficulties of users who explore EV market data.

Empathy Area	User Observations
Says	“I want to compare EV manufacturers.” “I need to understand which vehicle categories have higher recorded sales.” “I want to compare battery technology, driving range, and price.”

Empathy Area	User Observations
Thinks	Which manufacturers have higher recorded sales? Which battery types appear most frequently? Do prices vary with driving range? How do ratings differ among vehicles?
Does	Opens the dataset, examines columns, compares vehicle characteristics, applies filters, and reviews charts to identify patterns.
Feels	May feel overwhelmed by the size of the dataset and uncertain about which fields are most useful for comparison.
Pains	Large numbers of records, multiple dimensions, possible missing values, complex comparisons, and the risk of misleading conclusions.
Gains	Clear charts, concise KPIs, interactive filters, consistent measures, and a summary of observations supported by the dataset.

Design implication: The empathy map supports a single dashboard with clear sections, readable labels, consistent units, useful filters, and tooltips. The dashboard should focus on the most relevant comparisons instead of displaying every field at once.

2.3 Brainstorming

During the ideation phase, different approaches were considered for analysing the dataset.

S. No.	Proposed Idea	Category	Priority
1	Compare annual sales by manufacturer.	Sales Analytics	High
2	Compare sales across vehicle categories.	Market Analysis	High
3	Visualise the distribution of battery technologies.	Battery Analysis	High
4	Compare driving range and energy efficiency.	Performance Analysis	High
5	Analyse the relationship between price and driving range.	Pricing Analysis	High
6	Compare charging power and charging times.	Charging Analysis	Medium
7	Compare customer and safety ratings.	Customer Insights	Medium
8	Analyse manufacturing countries, regions, and model years.	Geographic Analysis	Medium
9	Explore CO_2 savings and recommended vehicle use cases.	Sustainability Analysis	Medium
10	Develop a machine-learning model for future EV sales prediction.	Predictive Analytics	Low for current scope

Final selected idea: An interactive EV Market Analysis Dashboard that combines the most relevant descriptive analyses in one place. This idea is feasible because the supplied dataset already contains the fields needed for the proposed visualisations. The current project scope emphasises descriptive analytics rather than advanced machine learning.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The customer journey describes how a user interacts with the EV Market Analysis Dashboard.

Stage	User Activity	Possible Difficulty	Dashboard Support
1. Access	Opens the dashboard.	May not know where to begin.	Clear title and overview.
2. Understand	Reviews summary KPIs.	Figures may lack context.	KPI cards with labels and units.
3. Explore	Examines manufacturer and category sales.	Manual comparison is difficult.	Sorted charts and category comparisons.
4. Filter	Selects a manufacturer, battery type, country, or year.	Too much data may remain visible.	Interactive filters and slicers.

Stage	User Activity	Possible Difficulty	Dashboard Support
5. Compare	Examines price, range, efficiency, and ratings.	Measures use different units.	Clearly labelled axes and tooltips.
6. Interpret	Reviews observed patterns.	Charts may be misinterpreted.	Summary insights and data limitations.
7. Present	Shares findings in a project demonstration.	Results may be difficult to explain.	Organised layout and clear views.

The journey map guides the arrangement of dashboard content from a general overview to more detailed comparisons.

3.2 Solution Requirement

A. Functional Requirements

- Load and display the EV dataset.
- Present summary KPIs for relevant measures.
- Compare annual sales across manufacturers and vehicle categories.
- Display battery-type distribution.
- Compare driving range and energy efficiency.
- Explore vehicle price in relation to driving range.
- Compare charging power and charging time.
- Analyse customer ratings and safety ratings.
- Explore manufacturing countries, market regions, and model years.
- Display sustainability and recommended-use-case comparisons where appropriate.
- Provide interactive filters for relevant categorical fields.
- Display clear chart titles, legends, tooltips, and units.

These are intended functions. The final report should list only those confirmed to exist in the completed workbook as implemented features.

B. Non-Functional Requirements

Requirement	Description
Usability	The dashboard should be understandable to users with basic data-analysis knowledge.
Performance	Visualisations and filters should respond within an acceptable time on the available system.
Accuracy	Aggregated measures should be validated against the source dataset.
Readability	Titles, labels, legends, and units should be clearly visible.
Consistency	Similar categories and measures should use consistent naming and formatting.
Reliability	The dashboard should display correct results after filter changes.
Maintainability	The dataset and dashboard should be organized so that updates can be managed.
Privacy	Publishing should be performed only when the dataset is suitable for public sharing.

C. Hardware and Software Requirements

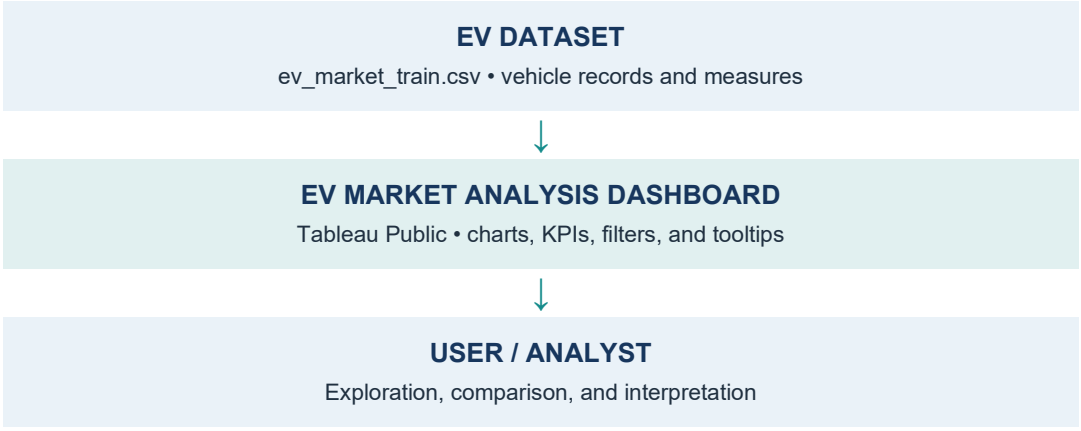
Hardware: a computer or laptop capable of running a modern browser, sufficient memory to work with a large dataset, and a stable internet connection for Tableau Public and online resources.

Software: Windows, macOS, or another supported operating system; a current browser such as Google Chrome or Microsoft Edge; Tableau Public; and the ev_market_train.csv data source.

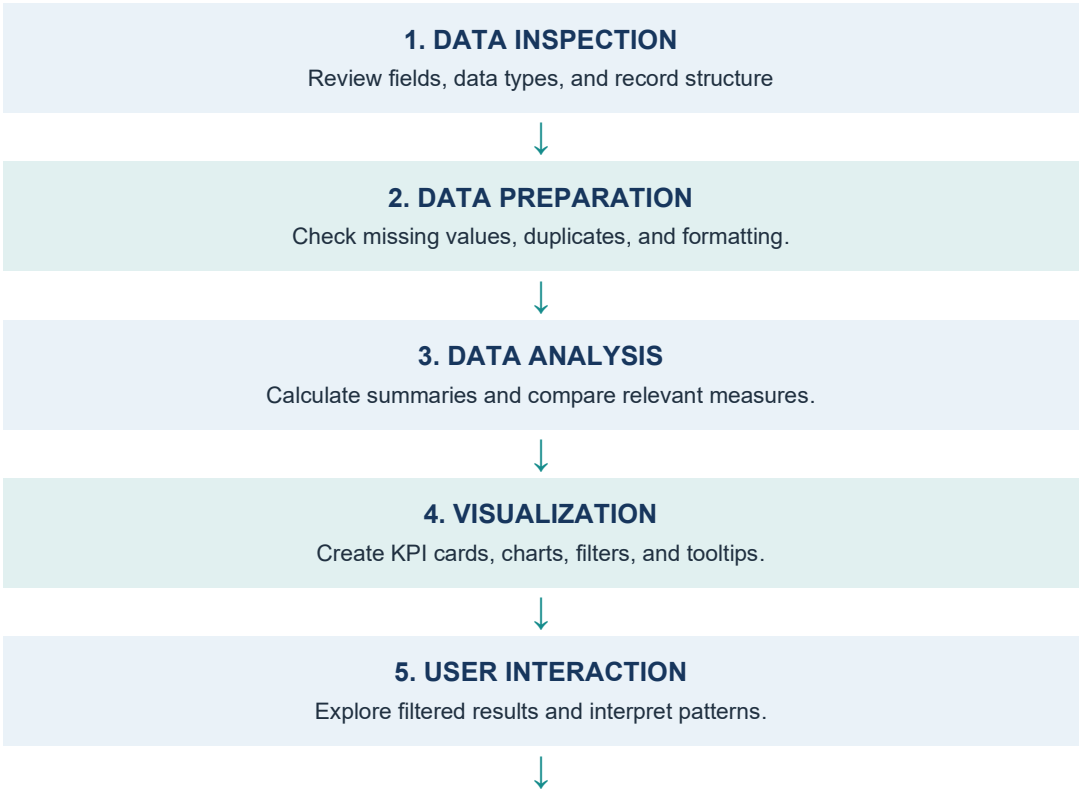
Python or SQL may be used for additional preparation or validation only if they are part of the actual implementation.

3.3 Data Flow Diagram

Level 0: Context Diagram



Level 1: Data Processing Diagram



The diagrams represent the intended analytical workflow. If the actual implementation uses only Tableau Public, the report should not imply that separate Snowflake or Python processing components have been deployed.

3.4 Technology Stack

Technology	Purpose
Tableau Public	Creating interactive dashboards, visualisations, filters, and stories.
CSV dataset	Providing the EV records and attributes for analysis.
Web browser	Viewing and interacting with the dashboard.
Spreadsheet software (optional)	Inspecting columns, checking sample records, and preliminary validation.
SQL (optional)	Performing additional data-quality checks and aggregation validation if a database is used.
Python/Pandas (optional)	Performing programmatic data profiling or cleaning if included in the implementation.

The core visualisation tool for this documentation is Tableau Public. Optional technologies should be retained in the final report only if they were actually used.

4. PROJECT DESIGN

4.1 Problem-Solution Fit

User Problem	Proposed Solution	Expected Benefit
Comparing many manufacturers manually is time-consuming.	A sorted manufacturer sales chart.	Easier comparison of recorded sales.
Vehicle categories are difficult to compare in a raw table.	Category-level charts.	Clearer category comparisons.
Battery and performance attributes are spread across many fields.	Battery, range, and efficiency visualisations.	Easier examination of technical characteristics.
Price and range are difficult to compare simultaneously.	A price-versus-range scatter plot.	Better visibility of patterns and outliers.
Market differences are difficult to investigate.	Geographic and model-year visualisations.	Easier exploration by location and year.
Static summaries cannot answer every user question.	Filters and tooltips.	More flexible exploration.

The dashboard aims to reduce the effort needed to explore the dataset. Its usefulness depends on correct calculations, clear visual design, and appropriate handling of missing or unusual values.

4.2 Proposed Solution

The project solution is an interactive dashboard that presents EV market information through a set of organised visualisations.

KPI Cards: Possible KPIs include total recorded annual sales, average driving range in kilometres, average vehicle price in USD, average customer rating, and the number of vehicle records represented after filtering.

Manufacturer Sales: A horizontal bar chart can compare recorded annual sales by manufacturer.

Category Sales: A sorted bar chart can compare sales totals across vehicle categories.

Battery Analysis: A bar chart or similar visualisation can show the distribution of battery types. Record count and sales volume must be distinguished clearly.

Range and Efficiency: Charts can compare driving range and energy efficiency by manufacturer or category.

Price Versus Range: A scatter plot can show the relationship between vehicle price and range. This helps reveal patterns and outliers but does not prove that price causes longer range.

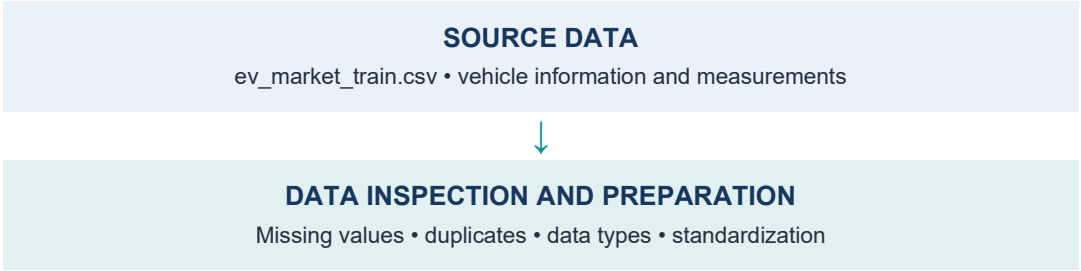
Charging Performance: Charts can compare charging power and charging times.

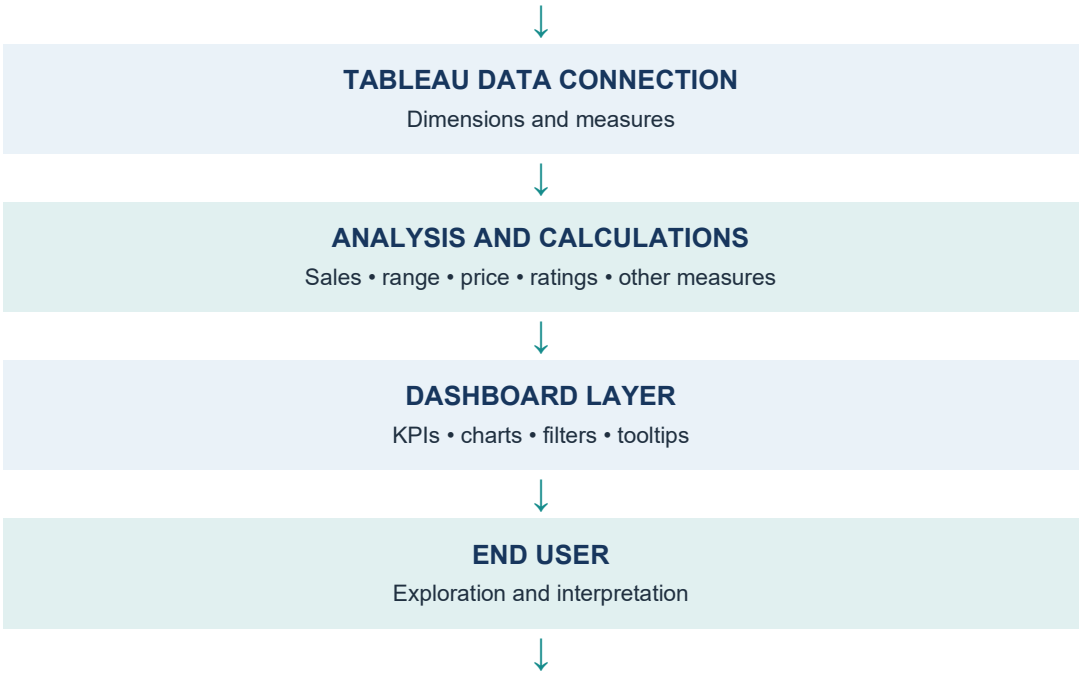
Customer and Safety Ratings: Visualisations can compare available customer ratings and safety ratings.

Geography and Sustainability: Country, region, model-year, CO_2 savings, and recommended-use-case fields can support further descriptive analysis.

The proposed filters include manufacturer, category, battery type, manufacturing country, market region, model year, drivetrain, and recommended use case. Tooltips can display additional details when users inspect a data point. The final set of visualizations and controls should match the actual workbook.

4.3 Solution Architecture





The architecture separates source data, preparation, calculations, visualization, and user interaction. If no separate preparation tool was used, the data-preparation stage may be carried out through available Tableau functionality and should be documented accordingly

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5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project can be managed through a sequence of phases, from data understanding to final dashboard validation.

Phase	Main Activities	Deliverables
1. Ideation	Define the project problem, target users, and dashboard concept.	Problem statement and ideation notes.
2. Requirement Analysis	Identify user needs, functional requirements, data fields, and technology.	Requirements and design diagrams.
3. Data Preparation	Inspect records, assess data quality, and document treatment decisions.	Prepared or validated dataset.
4. Dashboard Design	Plan KPIs, visualizations, colour scheme, and layout.	Dashboard design and visualization plan.
5. Development	Connect the dataset and create charts, calculations, filters, and layouts.	Tableau workbook.
6. Testing	Validate aggregations, filters, labels, and responsiveness.	Testing checklist and screenshots.
7. Documentation	Prepare the report and collect supporting evidence.	Final project documentation.
8. Submission	Verify project files, links, screenshots, and presentation readiness.	Submission package.

Suggested Scheduling Plan

Task	Estimated Duration
Ideation and requirement analysis	1–2 days
Data inspection and preparation	1–2 days
Dashboard design	1–2 days

Task	Estimated Duration
Visualization and dashboard development	3–5 days
Testing and corrections	1–2 days
Documentation and final review	1–2 days

These durations are planning estimates rather than a record of actual time spent. The final report may replace them with the team's actual schedule.

Expected deliverables include the EV dataset and data-quality notes, Tableau workbook or published dashboard, interactive charts and filters, dashboard screenshots, functional and performance testing evidence, final report, source code or calculated fields where applicable, and GitHub/demo links if available.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Testing verifies that the dashboard loads correctly, displays expected data, calculates measures properly, and responds to user interactions. Performance results should be based on actual tests. Execution times, loading times, and the number of completed visualizations must not be estimated and then presented as measured results.

A. Functional Testing

S. No.	Test Case	Expected Result	Actual Result
1	Open the dashboard.	Dashboard loads successfully.	To be recorded.
2	Verify the dataset connection.	Correct data source and fields are available.	To be recorded.
3	Check KPI calculations.	KPI values match validated dataset calculations.	To be recorded.
4	Apply a manufacturer filter.	Relevant charts update according to the selection.	To be recorded.
5	Apply a category filter.	Category-dependent visualizations update correctly.	To be recorded.
6	Apply a model-year filter.	Displayed results reflect the selected year or range.	To be recorded.
7	Inspect chart tooltips.	Tooltips display the correct measures and dimensions.	To be recorded.
8	Compare chart totals with source data.	Aggregated values agree with validation calculations.	To be recorded.

B. Performance Testing Template

S. No.	Parameter	Screenshot / Values
1	Data Rendered	Previously inspected dataset: 200,000 records and 37 columns. Attach the actual Tableau data-source screenshot and record loading time if measured.
2	Data Preprocessing	Document missing values, duplicates, inconsistent labels, and cleaning performed. Attach evidence of actual checks.
3	Utilization of Filters	Test manufacturer, category, battery type, country, region, and model-year filters. Record response observations and attach screenshots.
4	Calculation Fields Used	Record the calculated fields that exist in the workbook and validate their values.
5	Dashboard Design	Record the actual number of completed visualizations and KPI cards. Attach the dashboard screenshot.
6	Story Design	Record the actual number of Tableau Story points, if a Story was created. Attach a screenshot.

C. Performance Measurements

Metric	Measurement
Dashboard initial load time	Record actual measured time.
Filter response time	Record actual measured time for representative filters.
Data completeness	Calculate from actual missing-value checks.
Duplicate records	Record the result of duplicate analysis.
Calculation accuracy	Compare dashboard measures with validated totals or averages.
Visualization count	Count actual completed charts.
Story point count	Count completed Story points, or mark “Not applicable” if no Story was created.

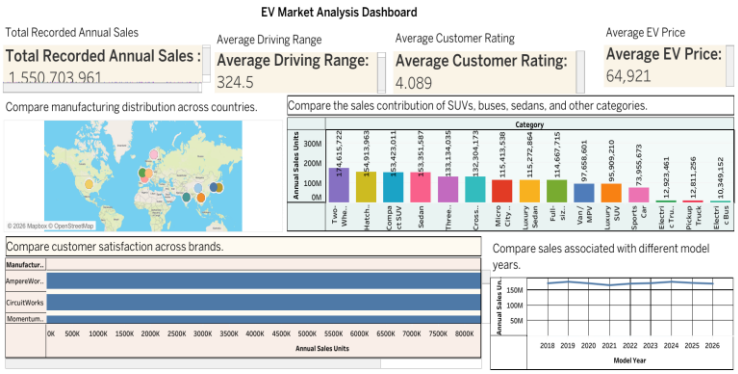
Testing outcome: Testing is complete only when actual results have been recorded and reviewed. The final testing report should distinguish successful tests from failures, pending tests, and tests that were not applicable.

7. RESULTS

7.1 Output Screenshots

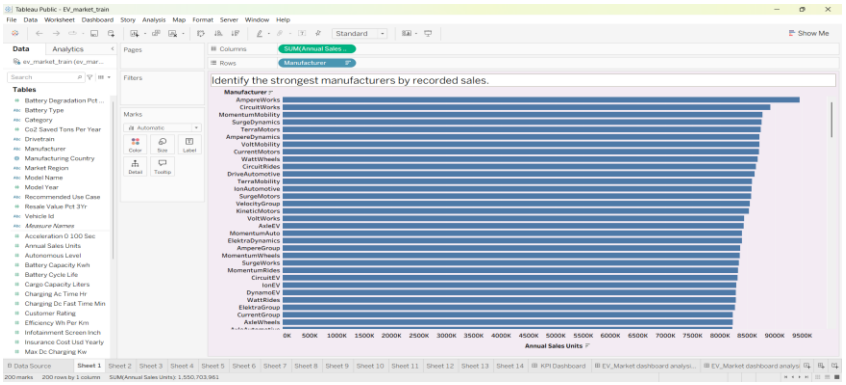
This section should contain evidence of the actual project implementation. Add screenshots from the completed Tableau workbook rather than illustrations or proposed layouts.

Screenshot 1: Dashboard Overview



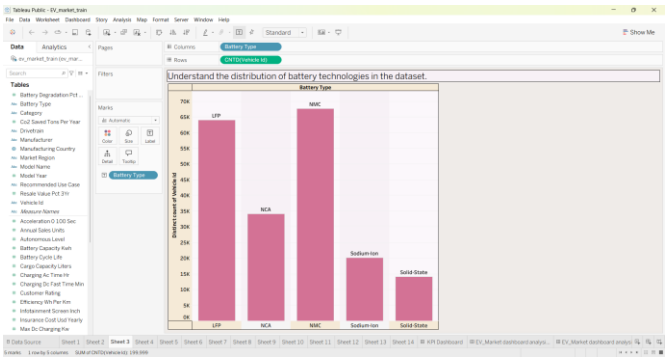
Show the dashboard title, main KPIs, visualization layout, and available filters. Description: demonstrates the overall organization and main information available to users.

Screenshot 2: Manufacturer Sales Analysis



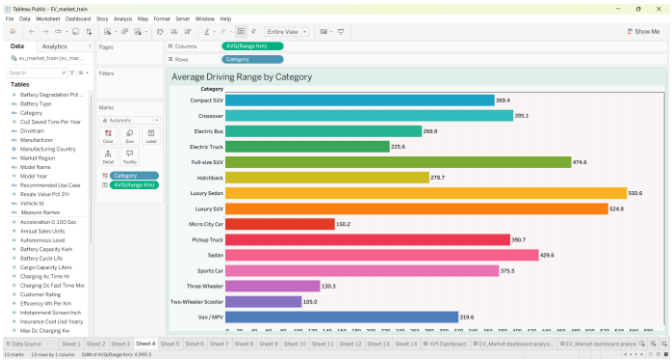
Show the manufacturer sales chart. Description: compares recorded annual sales across manufacturers.

Screenshot 3: Vehicle Category and Battery Analysis



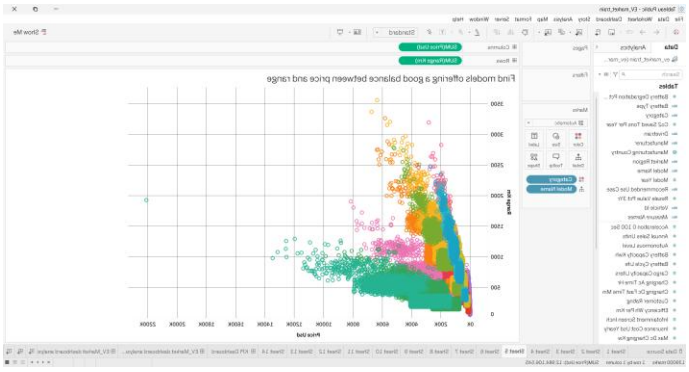
Show the category comparisons and battery-type distribution. Record counts should not be confused with sales totals.

Screenshot 4: Price and Range Analysis



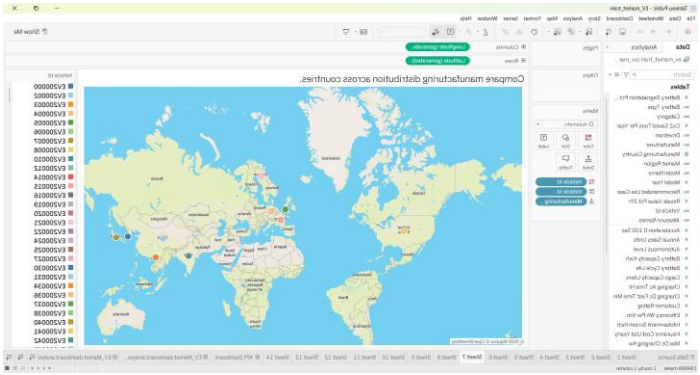
Show the price-versus-range visualization. Description: supports investigation of patterns and outliers.

Screenshot 5: Interactive Filters



Show a selected filter and the resulting dashboard update.

Screenshot 6: Customer Ratings or Geographic Analysis



Show customer ratings, manufacturing countries, market regions, or model-year differences.

Results Summary

The project is intended to make the EV dataset easier to explore by consolidating relevant measures into interactive visualizations. The completed results should be evaluated using the actual dashboard, validated aggregations, and recorded filter behaviour. Specific findings, such as the manufacturer with the highest recorded sales or the most frequent battery type, should be inserted only after verifying them from the final data and dashboard.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Easy interpretation: charts provide a visual summary of a large dataset.
- Interactive analysis: users can filter the dataset by relevant dimensions.
- Multiple comparisons: sales, technical specifications, pricing, ratings, and geography can be explored in one project.
- Time saving: users need not manually compare every record in a spreadsheet.
- Clear presentation: visualizations help communicate descriptive findings to reviewers and stakeholders.
- Extensible design: additional views can be added if suitable data and project time are available.
- Accessibility: Tableau Public allows a dashboard to be shared online when public publication is appropriate.

Disadvantages and Limitations

- Dependence on data quality: missing or inconsistent source values can affect calculations.
- Limited representativeness: the dataset may not reflect the entire global EV market.
- No causal conclusions: observed relationships between price, range, sales, or ratings do not prove causation.
- Scope limitations: the current project focuses on descriptive analysis rather than forecasting.
- Performance variation: dashboard responsiveness can depend on data volume, chart complexity, and the user's system or connection.
- Public visibility: Tableau Public workbooks and published data may be accessible to others.
- Interpretation risks: incorrect aggregation or unclear labels can lead to misleading conclusions.

9. CONCLUSION

The EV Market Analysis Dashboard project demonstrates how data visualization can be used to explore a large dataset containing vehicle, sales, technical, pricing, rating, geographic, and sustainability information.

The ev_market_train.csv dataset provides multiple dimensions and measures for descriptive analysis. Tableau Public can present these attributes through charts, KPI cards, filters, and tooltips, helping users compare manufacturers and categories, examine battery technology, investigate driving range and pricing, and explore recorded ratings and sales patterns.

The project also highlights the importance of data preparation and validation. Missing values, duplicate records, inconsistent labels, invalid values, and aggregation choices must be considered before drawing conclusions.

Overall, the project provides a structured approach to EV market exploration. Its final quality depends on the accuracy of the calculations, the clarity of the dashboard, and the evidence collected during testing.

10. FUTURE SCOPE

Sales forecasting: Add a predictive model when suitable historical and representative data are available.

Market share analysis: Integrate reliable total-market sales data and definitions to calculate market share appropriately.

Expanded data sources: Add verified information from additional EV manufacturers, regions, and years.

Advanced filters: Add detailed comparisons by drivetrain, use case, battery capacity, or vehicle specifications.

Deeper battery analysis: Explore battery capacity, cycle life, degradation, and charging performance together.

Improved performance: Simplify complex views and optimize data structure if dashboard responsiveness becomes an issue.

Regular updates: Establish a repeatable workflow for loading and validating newer records.

Accessibility improvements: Improve colour contrast, labels, and navigation for different users.

Additional analytical methods: Explore clustering or statistical analysis if the project scope and data quality support them.

Enhanced reporting: Provide verified summary insights and downloadable reports where appropriate.

These enhancements are possible future developments and should not be described as existing functionality unless implemented.

11. APPENDIX

A. Source Code / Calculated Fields

The main deliverable is a Tableau Public dashboard. Tableau workbooks typically contain visualization configuration and calculated fields rather than a conventional application source-code file. Include each field below in the final appendix only if it exists in the actual workbook.

A.1 Total Recorded Sales

```
SUM([annual_sales_units])
```

Purpose: Aggregates recorded annual sales values for the selected records. Label this as recorded sales and validate it against the dataset.

A.2 Average Driving Range

```
AVG([range_km])
```

Purpose: Calculates the average available driving range within the selected records.

A.3 Average Vehicle Price

```
AVG([price_usd])
```

Purpose: Calculates the average available recorded vehicle price.

A.4 Price per Kilometre of Range

```
IF [range_km] > 0 THEN
    [price_usd] / [range_km]
END
```

Purpose: Calculates price divided by driving range when the range is positive and available. It is a derived comparison measure, not a complete assessment of vehicle value.

A.5 Range Band

```
IF ISNULL([range_km]) OR [range_km] < 0 THEN
    "Unknown"
ELSEIF [range_km] < 250 THEN
    "Short Range"
ELSEIF [range_km] <= 400 THEN
    "Medium Range"
ELSE
    "Long Range"
END
```

Purpose: Creates user-defined groups for comparing vehicles by range. These thresholds are project-specific, not official industry classifications.

A.6 Price Band

```
IF ISNULL([price_usd]) OR [price_usd] <= 0 THEN
    "Unknown"
ELSEIF [price_usd] < 30000 THEN
    "Budget"
ELSEIF [price_usd] <= 60000 THEN
    "Mid-Range"
ELSE
    "Premium"
END
```

Purpose: Groups records into project-defined price ranges to simplify visual comparisons.

B. Dataset Link

Item	Details
Dataset filename	ev_market_train.csv
Previously inspected size	200,000 records and 37 columns
Original dataset URL	https://1drv.ms/x/c/2b7fd3d8207dac7f/IQBocQXZ6neOTp3hJlPAT652AZwSksrPYbF5suZe769Fo1U?e=w6koQV

The dataset was supplied as a project file. If it has no public URL, state that it was provided as a local project dataset rather than inventing an external source.

C. GitHub Link

GitHub Repository:: https://github.com/ankitaroy1902/Electric-Vehicle-Market-Performance-Insights.git

D. Project Demo Link

Tableau Public Dashboard: [https://public.tableau.com/views/EV_market_train/Story1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link].

If the dashboard or video is not publicly available, leave the relevant field blank or mark it as not available.

FINAL DOCUMENTATION NOTE

Replace all placeholders with the corresponding project details. Confirm dataset statistics against the exact file used in the final workbook, remove unimplemented features from implemented-results sections, and include only genuine screenshots and verified test results.